Kannuru Srinadh

Doctoral Fellow (PhD)
Adaptive System Lab (ASL) & ISDR & Comm. Net. Lab
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Research Interests

- **Sparse Signal Processing:** Techniques for efficient representation and processing of signals.
- **Model-Based Deep Learning:** Integrating traditional model-based approaches with deep learning frameworks.
- **Multimodal Image Super-Resolution:** Enhancing image resolution using data from multiple modalities (e.g., RGB–NIR, RGB–Thermal).
- **Deep Learning & Machine Learning Architectures:** Developing and optimizing architectures for various AI applications.
- **Computer Vision & Object Detection:** Algorithms for object detection and image analysis.
- **Biomedical Image Processing:** Applying image processing techniques to medical imaging for diagnosis and analysis.

Education

Doctor of Philosophy (PhD) in Electronics and Communication Engineering *July 2022 – Expected July 2027*
National Institute of Technology Rourkela CGPA: 8.58
– **Thesis Title:** Sparse Enabled Model based Deep Learning for Multimodal Image Superresolution
– **Supervisor(s):** Prof. Upendra Kumar Sahoo & Prof. Santos Kumar Das
– **Specialization:** Signal and Image Processing.

Master of Technology (M.Tech) in Electronics and Communication Engineering *Aug 2014 – Oct 2016*
Jawaharlal Nehru Technological University Kakinada CGPA: 8.7
– **Specialization:** Systems and Signal Processing.

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering *Aug 2009 – July 2013*
Jawaharlal Nehru Technological University Kakinada Percentage: 70.64 %

Intermediate (10+2) *June 2007 – March 2009*
Margadarsi Junior College, Gajuwaka, Visakhapatnam Percentage: 85.6 %

Class X *July 2006 – March 2007*
Z.P.H School, Vada cheepurupalli, Visakhapatnam Percentage: 77 %

Research Experience

Doctoral Fellow (PhD) *July 2022 – Present*
National Institute of Technology Rourkela Supervisor: Prof. Upendra Kumar Sahoo & Prof. Santos Kumar Das
– Conducting research on Sparse-Enabled Model-Based Deep Learning for Multimodal Image Super-Resolution.
– Developed novel deep learning architectures for multimodal sensor fusion.
– Implementing sparse representation techniques to enhance feature extraction and reconstruction.
– Analyzing and optimizing model-based frameworks for improved super-resolution performance.

Publications

- P. K. Sahu, S. Baraha, U. K. Sahoo, **K. Srinadh**, and S. K. Das. (**2027**). Distributed estimation of convex event regions in wireless sensor networks using diffusion cooperation and minimum volume ellipsoids. *Ad Hoc Networks, vol. 194, 104404.* DOI: 10.1016/j.adhoc.2026.104404.
- V. Kadha, B. B. Duddeti, **K. Srinadh**, S. K. Buddepu, L. Janjanam, and K. Medhi. (**2026**). From pixels to text: A deep learning survey of scene text detection and recognition. *Computers and Electrical Engineering, vol. 135, 111139.* DOI: 10.1016/j.compeleceng.2026.111139.
- **K. Srinadh**. (**2026**). Task-Aware Super-Resolution for Brain MRI: An Evaluation Framework Based on Downstream Segmentation Utility. *Computerized Medical Imaging and Graphics.* **(Submitted)**.
- A. Kumar, Y. D. Kumar, **K. Srinadh**, and U. K. Sahoo. (**2026**). UCRS: Uncertainty-Calibrated Region Sampling for Real-Time Small Object Detection in High-Resolution Images. *Proceedings of the National Conference on Communications (NCC), Hyderabad, India, pp. 418–423.* DOI: 10.1109/NCC68160.2026.11479125.

- **K. Srinadh**, K. Ramakrishna, V. David, and G. Lakshmi. (**2026**). HAD-SR: Hybrid Attention-Augmented Dictionary Learning for Unpaired Image Super-Resolution. *Proceedings of the National Conference on Communications (NCC), Hyderabad, India, pp. 84–89.* DOI: 10.1109/NCC68160.2026.11479114.
- **K. Srinadh**, K. Ramakrishna, V. David, and G. Lakshmi. (**2025**). Dual Multiplicative Intrinsic Components Optimization for Enhanced MRI Bias Field Correction. *Proceedings of the 6th International Conference on Recent Advances in Information Technology (RAIT), Dhanbad, India, pp. 1–6.* DOI: 10.1109/RAIT65068.2025.11089089.
- S. Rana, **K. Srinadh**, K. Bhagyasekhar, V. K. Kappala, and S. K. Das. (**2025**). Performance Analysis of FSO Communication Under Foggy Conditions. *Proceedings of the 4th International Conference on Range Technology (ICORT), Chandipur, India, pp. 1–6.* DOI: 10.1109/ICORT64008.2025.11115410 **(Best Paper Award)**.
- **K. Srinadh**, V. David, G. Lakshmi, and K. Ramakrishna. (**2024**). Unified Dictionary Training Approach for High-Resolution Image Enhancement. *Proceedings of the 4th International Conference on Artificial Intelligence and Signal Processing (AISP), Vijayawada, India, pp. 1–5.* DOI: 10.1109/AISP61711.2024.10870610.
- **K. Srinadh**, Y. D. Kumar, U. Sahoo, and S. K. Das. (**2026**). QSRNet: Hybrid Quantum-Classical Learning with Variational Circuits for Enhanced Feature Representation in Image Super-Resolution. *Proceedings of the Signal Processing and Communications Engineering (SPACE).* **(Accepted)**.
- Y. D. Kumar, **K. Srinadh**, and U. Sahoo. (**2026**). Hypergraph-Enhanced Lightweight Detection Architecture for Foreign Object Debris Identification on Airport Runways. *Proceedings of the Signal Processing and Communications Engineering (SPACE).* **(Accepted)**.
- C. B. Deepthi, **K. Srinadh**, Y. D. Kumar, and U. Sahoo. (**2026**). From Detection to Tracking: Benchmarking YOLOv8 to YOLOv26 for Real-Time UAV Surveillance. *Proceedings of the Signal Processing and Communications Engineering (SPACE).* **(Accepted)**.

Projects

Multimodal Image Super-Resolution using Deep Learning *2023 – Present*

Technologies: Python, TensorFlow, PyTorch

– Developing novel sparse-enabled model-based deep learning approaches for enhancing image resolution using multiple modalities.

IoT-based Smart Monitoring System *2022 – 2023*

Technologies: Arduino, Raspberry Pi, MQTT, Node.js

– Designed and implemented an IoT-enabled monitoring system for real-time data collection and analysis.

Signal Processing for Wireless Communications *2021 – 2022*

Technologies: MATLAB, C++

– Developed adaptive signal processing algorithms for improving wireless communication performance.

Machine Learning for Network Optimization *2020 – 2021*

Technologies: Python, scikit-learn, NumPy

– Applied machine learning techniques to optimize network resource allocation and improve QoS.

Computer Vision for Object Detection *2019 – 2020*

Technologies: TensorFlow, OpenCV

– Implemented deep learning models for real-time object detection and classification using CNN architectures.

Teaching Experience

Teaching Assistant *July 2022 – Present*

National Institute of Technology, Rourkela

– Course: Advanced Wireless Communications.
– Conducted weekly tutorial sessions for 40+ undergraduate students on MIMO, OFDM, and channel coding.
– Assisted in developing lab exercises and grading assignments.
– Provided one-on-one mentorship to students.

Assistant Professor *Sep 2017 – July 2022 (4.10 years)*

Anurag Engineering College (A), Telangana

– Taught multiple courses in Electronics and Communication Engineering.
– Conducted practical laboratory sessions.
– Mentored undergraduate students in their projects.

Courses Taught

- **Theory Courses:** Basic Electronics Engineering, Digital Electronics, Signals and Systems, Analog Communication, Digital Communication, Digital Signal Processing, Digital Image Processing, Information Theory and Coding, Adaptive Signal Processing, Advanced Digital Signal Processing.

- **Practical/Lab Courses:** AI and ML Laboratory, Basic Electronics Laboratory, Communication Engineering Laboratory, Digital Signal Processing Laboratory, Digital Communication Laboratory, Analog Communication Laboratory, Advanced Digital Signal Processing Laboratory, Adaptive Signal Processing Laboratory.

Awards and Honors

- **Best Paper Award** International Conference on Range Technology (ICORT 2025) for the paper "Performance Analysis of FSO Communication under Foggy Conditions." Conducted by Integrated Test Range (ITR) DRDO, Ministry of Defence, Government of India, Chandipur.
- **NIT Rourkela Research Fellowship** Awarded for outstanding academic performance and research potential during PhD studies (2022–Present).

Professional Service & Affiliations

- **Reviewer** IEEE International Conference on Acoustics, Speech, and Signal Processing (2025–Present)
- **Reviewer** Elsevier *Ad Hoc Networks* (2023–Present)
- **Organizing Committee Member** National Conference on Advanced Computing and Communication (NCACC 2023), NIT Rourkela (2023)
- **Membership** IEEE Council on RFID and IEEE Sensor Council
- **Student Member** Association for Computing Machinery (ACM)

Technical Skills

Programming Languages: Python, C/C++, JavaScript
Web Technologies: HTML5, CSS3, React
ML/AI Frameworks: TensorFlow, PyTorch, NumPy, scikit-learn, OpenCV
Tools & Platforms: MATLAB, Git/GitHub, MySQL, Arduino, Raspberry Pi, MQTT, Node.js
Other Tools: VS Code, LaTeX, Overleaf
Operating Systems: Linux, Windows
Areas of Expertise: Computer Vision, Deep Learning, Image Super-Resolution, Sparse Signal Processing, Model-Based Deep Learning, Image Enhancement, Biomedical Image Processing

Certifications

NIT Rourkela: Short Course on Computational Engineering with AI/ML (March 16–20, 2026).

IIT Hyderabad: NCC 2026 - Communications (Feb 26–March 1, 2026).

NIT Rourkela: Short-Term Course on Radar Signal & Image Processing (Feb 23–27, 2026).

NIT Rourkela: Short Course on ACCT-CPS 2026 (January 8–12, 2026). Certificate ID: CT-25-CS-053.

Isha Foundation: Inner Engineering Program (January 5, 2026). Certificate ID: CERT001380.

NIT Rourkela: Online Short-Term Course & Faculty Development Program on Innovation in Control and Learning for Dynamical Systems (ICLDS-2024) (Nov 29–Dec 03, 2024).

NIT Rourkela: National Workshop on AI for Digital Healthcare Innovations (AIDHI-2024) (October 15–19, 2024). Certificate ID: CT/24/CS/049/031.

NIT Rourkela: Short-Term Course on Exploring Deep Learning Applications and Practical Implementations (June 11–15, 2024). Certificate ID: NITR/CS/2024/M/0892/0126.

NIT Rourkela: Short-Term Course on Recent Trends in Computer Vision (December 11–15, 2023). Certificate ID: NITR/CS/2024/M/0149/012.

NIT Rourkela: Workshop on Shodh Shala on Research Methodology and Data Analysis (March 1–5, 2023).

VIT Chennai: Advanced Electronic Devices (2023).

NIT Warangal: Five-Day Short-Term Training Program on "Data Science for All with R" (Online Mode) (Feb 22–26, 2023).

NIT Rourkela: National Conference on Advanced Computing and Communication (NCACC 2023).

Anurag University: Faculty Development Program on Internet of Things (IoT) (2020).

Gudlavalleru Engineering College: Electric Vehicle Technologies (June 2020).

CMR Engineering College: Faculty Development Program on Artificial Intelligence (May 2020).

Narasaraopeta Engineering College: FDP on OBE – NBA Process (2020).

NAAC: Accreditation Framework of NAAC (2020).

Anurag Group of Institutions: Research to Publication Workshop (2020).

SV College of Engineering: Career Guidance in VLSI (2020).

MMIT: NAAC Awareness Program (May 2020).

University of Michigan (Coursera): HTML5 Programming (2020).
Google (Coursera via Rhyme): Informative Presentations (2020).
Webinar Certification: Web Tools for Research (2022).
Anurag University: Professional Development Certificate (2020).
Swayam: Computer Fundamentals (November 2019).
Professional Assessment: Digital Communications Technical Quiz (2020).

Professional Development & Service

Tech. Workshop Coordinator"Python for Beginners" Workshop – NIT Rourkela *April 2025*

- Organized and taught a 2-day hands-on lab for 50+ students, covering fundamental Python concepts and practical applications.
- Managed all logistical aspects, from resource allocation to material preparation, ensuring a smooth learning experience.

References

Prof. Upendra Kumar Sahoo
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Prof. Santos Kumar Das
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Relationship: PhD Co-Supervisor

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